Probit Regression Function

Applications in Government & Politics

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April 2026

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Topic 1 | History

From Science to Government

- **James Tobin (1918–2002)**
  - Popularized Probit in Gov/Economics.
  - Served on JFK's Council of Economic Advisers (1961–1962).
  - Won Nobel Prize for financial market analysis and their relations to expenditure decisions, employment, production, and prices.
- **Key Innovations**
  - Popularized **Maximum Likelihood Estimation (MLE)**.
  - Formulated **Latent Variable Theory**: $Y^* = X\beta + \epsilon$.
- **The 1958 Landmark Paper**
  - *"Estimation of Relationships for Limited Dependent Variables"*

Topic 2 | Prerequisite Knowledge

Bernoulli Distribution Variance

Used for modeling binary outcomes. The variance is defined as $\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$

General Linearized Model

A *General Linearized Model* (GLM) is a model that applies a linear function to a dataset, with the general rules of mapping the input to the output y, and mapping it according to a **Link Function**.

Topic 2 | Prerequisite Knowledge

Gradient

A vector of partial derivatives. It is used in mapping the log likelihood function of the linear model. For this case, we use the weights as slopes and apply the transformation $X^T \cdot W$, where W is the weights matrix. This is used in solving for the **Hessian**

Hessian

A matrix of 2nd order partial derivatives. For this case, however, we take the negative Expected Value of the hessian, which simplifies the matrix math. Using the previously defined weights matrix, a formula for solving the Expected Hessian becomes (2):

$$X^T W X^T$$

Link Function

The link function is the name of the function that distinguishes what type of GLM a specific Linear Model is. For the case of probit, the inverse of the CDF of the normal distribution is used as the link function (also known as the probit function). For Logistic, the logit function is used, which is the inverse logistic function.

- Logit: $\eta(\mu) = \ln\left(\frac{\mu}{1-\mu}\right)$
- Probit: $\eta(\mu) = \Phi^{-1}(\mu)$

Topic 3 | Theoretical and Mathematic Explanation

- **Assumptions**:
  - Has all assumptions of a GLM, found in (4)
  - Assumes a Bernouli Distribution for the y data
- **Inputs**:
  - x: the design matrix, defined as having all 1s in the first column vector, with the rest being vectors for the observed values of the data
  - y: the target variable, assumed Bernouli Distribution
  - max_iter: maximum iterations to do
  - tolerance: the level that is closest to zero that decides if there has been a "convergence" to the $\hat{B}$
  - verbose: used for debugging, prints the steps and beta values
  - *note* some of these inputs are handled by glm differently, like the design matrix X is created by just adding a column of 1s to the dataframe passed in as x

Topic 3 | Theoretical and Mathematic Explanation

- **Function**:
  - initialize beta to be a vector of 0s, of length m (number of x columns)
  - for(i in max_iter)
    - get the linear predictor, eta
    - calculate the PDF and CDF of the predictor assuming the normal distribution, defined as $\hat{p}\hat{q}$
    - calculate the ratio of the CDF to the bernouli variance of the PDF $\hat{q}/(\hat{p}(1-\hat{p}))$
    - get a score vector with the previous ratio in the form ratio $\cdot\ (\hat{y}-\hat{p})$
    - calculate the gradient, g, defined by $X^T \cdot$ score
    - now, we calculate the Weight matrix of the form $\frac{\hat{q}}{\hat{p}(1-\hat{p})}$ note the bernouli distribution in the denominator
    - using the Weight vector, we calculate the negative expected Hessian, H, this is written as $-(X^T \cdot W)$
    - finally, a copy of the current beta iteration is made, and the next iteration of beta becomes $\hat{B} = \hat{B} - (H^{-1}g)$ where g is the gradient
    - then the tolerance check is done, checking the squared sum of $\hat{B}_{new} - \hat{B}_{old}$, or $||B_{new} - B_{old}||^2$ if this is less than our tolerance, we have converged

Topic 3 | Theoretical and Mathematic Explanation

- **Usage**:
  - Probit has many uses, but most of them are in econometrics and Government. The closest model in terms of use case is Logistic, as they both follow bernouli variables.
  - The main difference from Logisitc regression is that logistic regression uses the logistic distribution
  - This makes it useful in studying things like human behavior, psychology, economics, and finance.

Topic 3 | Theoretical and Mathematic Explanation

Solving the System

Optimization Methods

- There are 2 main methods of creating a model for probit regression:
  - 1. Newton-Raphson method We used a method similar to Newton Raphson, but rather than use the Hessian, the Expected Hessian was used, which is used for IRLS and is simpler to calculate.
  - 2. Iteratively Reweighted Least Squares / Fisher Scoring
- These methods can be shown to be mathematically the same(7)

Topic 3 | Theoretical and Mathematic Explanation

- we will now explore the exact implementation of Probit in greater detail, showing intermediate calculations for the various components
- There will be 5 main steps:
  - Linear Predictor
  - linear predictor -> Probability
  - Score Vector
  - Hessian
  - Newton-Raphson Update

Step 1: Linear Predictor

$$\eta_i = x_i^\top \beta$$

```
eta <- X %*% beta
```

The linear predictor is passed through a standard normal CDF to convert it to a probability

Step 2: Linear Predictor –> Probability

$$\hat{p}_i = \Phi(\eta_i) \quad \text{and} \quad \hat{q}_i = \phi(\eta_i)$$

```
phat <- pnorm(eta)
qhat <- dnorm(eta)
```

Using the linear predictor, $\hat{p}_i = \Phi(\eta_i)$ converts it into a probability between 0 and 1 for $Y_i = 1$, while $\hat{q}_i = \phi(\eta_i)$ indicates the rate of change of that probability with respect to the linear predictor at that point.

Step 3: Score Vector

$$s(\beta) = \sum_{i=1}^{N} \frac{\phi(x_i^\top \beta)}{\Phi(x_i^\top \beta)\left(1 - \Phi(x_i^\top \beta)\right)} \left(y_i - \Phi(x_i^\top \beta)\right) x_i$$

$$\frac{\phi(\eta_i)}{\Phi(\eta_i)(1 - \Phi(\eta_i))} \times (y_i - \Phi(\eta_i)) \times x_{ij}$$

The score function tells us in which direction should the coefficients β to increase the probability of observing the data (dependent variable) given the parameters (independent variables).

```
ratio <- qhat / (phat * (1 - phat))
score <- (y - phat) * ratio
g     <- t(X) %*% score
```

The Score Vector

The score function has three components.

Ratio: tells us how sensitive the model is at that point $\eta_i = x_i^\top \beta$. At the start of the estimation, the coefficients β are initialized to 0. At this point, the predicted probability of all events is 0.5, meaning the model adjusts to errors and is positioned to learn.

Residual: compares what actually happened $y_i = 0$ or 1 with the predicted probability. If it is positive, the model predicted too low. If it is negative, the model predicted too high.

Gradient: tells the model both the **direction** (increase or decrease a coefficient) and the **magnitude** of adjustment needed so that predicted probabilities better align with the observed data.

$$\eta_i = x_i^\top \beta$$

$$\frac{\partial \eta_i}{\partial \beta_j} = x_{ij}$$

Step 4: Hessian

The weights tell the model to focus on observations that are hardest to classify (those with predicted probabilities near 0.5 ex. initialization), and the Hessian uses this information to determine how large a step to take when updating the coefficients.

$$\mathbf{H} = -\mathbf{X}^\top \mathbf{W} \mathbf{X}$$

$$w_i = \frac{\phi(\eta_i)^2}{\Phi(\eta_i)(1 - \Phi(\eta_i))}$$

```
w <- (qhat^2) / (phat * (1 - phat))
H <- -t(X) %*% (X * as.vector(w))
```

Step 5: Newton-Raphson Update

$$\theta_{new} = \theta_{old} - [\text{Hessian}]^{-1} \times [\text{Gradient}]$$

$$\beta^{(t+1)} = \beta^{(t)} - \mathbf{H}(\beta^{(t)})^{-1} s(\beta^{(t)})$$

The algorithm looks at three things:

1. Where it is now $\beta^{(t)}$
2. Which direction improves the model? **The gradient**
3. How aggressive the move in the direction should be? **The Hessian**

The new coefficients represent the difference between the previous coefficients and the new curvature adjusted step.

```
beta <- beta - solve(H, g)
```

The computer is repeating the same update step over and over, each time improving the coefficients, until they stop changing.

Topic 4 | Implementation

```r
myprobit = function(X, y, max_iter, tolerance, verbose=FALSE){

  m = ncol(X)
  beta <- rep(0, m)

  for (i in 1:max_iter) {

    eta <- X %*% beta

    phat <- pnorm(eta)
    qhat <- dnorm(eta)

    ratio <- qhat / (phat * (1 - phat))
    score <- (y - phat) * ratio
    g <- t(X) %*% score

    w <- (qhat^2) / (phat * (1 - phat))
    H <- -t(X) %*% (X * as.vector(w))

    copy_beta = beta
    beta <- beta - solve(H, g)

    if (verbose){
      cat("Iteration", i, ":\n")
      j=1
      for (b in beta){
        cat("Beta", j, " = ", beta[j], "\n")
        j=j+1
      }
    }
    if ((sum(beta - copy_beta)^2) < tolerance) {
        return(beta)
        break
      }
  }
}
```

Verifying with R's in-built function, glm

```r
X <- model.matrix(am ~ mpg, data = mtcars)
y <- mtcars$am
myprobit(X,y,32, 0.00009, FALSE)
```

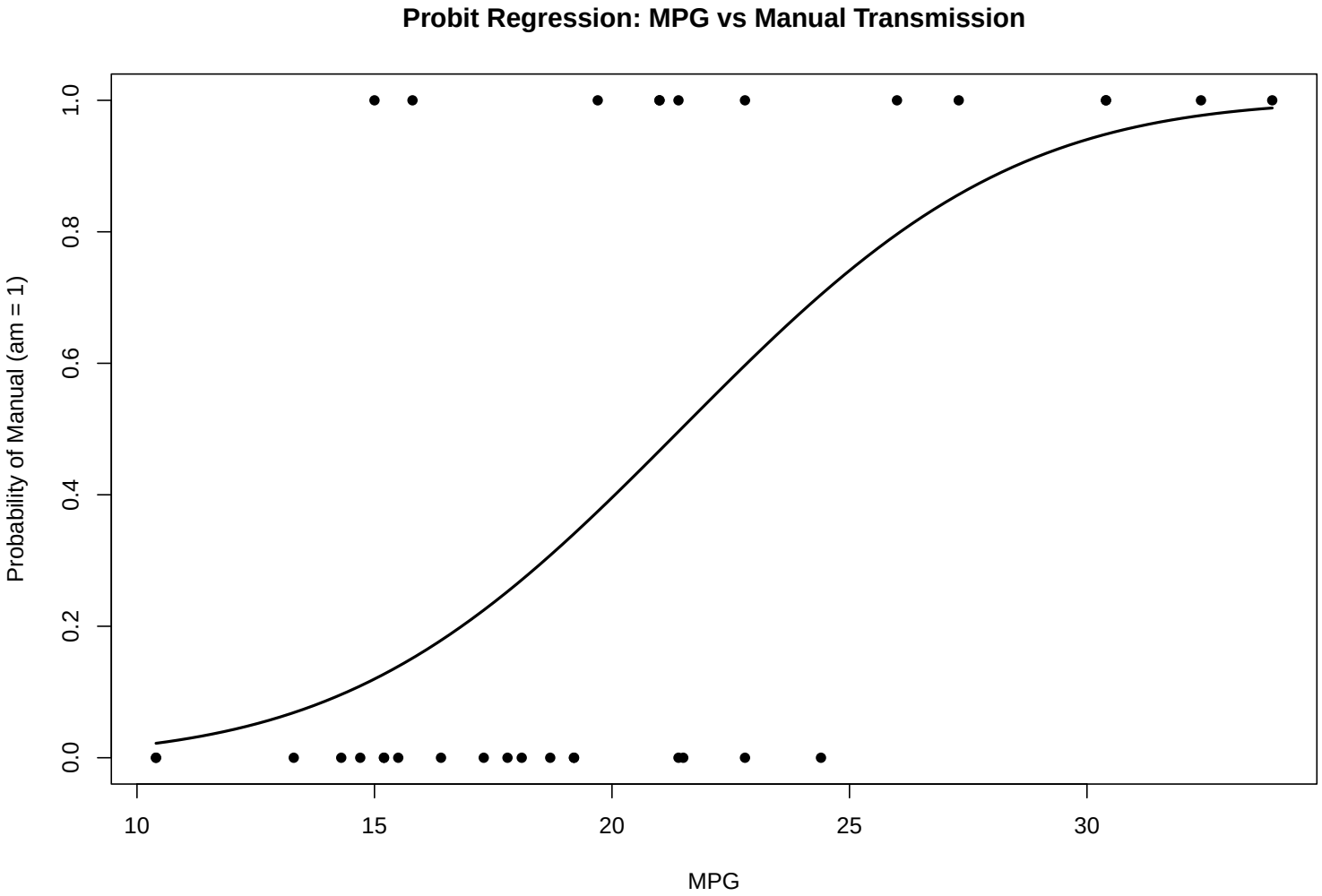
```
##                    [,1]
## (Intercept) -3.9100663
## mpg          0.1822605
```

```r
modell <- coef(glm(am ~ mpg, data = mtcars, family = binomial(link="probit")))
modell
```

```
(Intercept)         mpg
 -3.9102065   0.1822684
```

- Intercept **(-3.91)** represents the baseline latent tendency for a car to be manual when MPG is zero; it is strongly negative, meaning the model starts from a situation where automatic transmission is much more likely before considering MPG.
- The MPG coefficient **(0.182)** means that each additional unit of MPG increases the latent "manual likelihood" score, shifting the probability toward a manual transmission.

Initial Relationship Examination



There is a positive relationship between MPG and the probability that a car has a manual transmission: as fuel efficiency increases, the likelihood that a car is manual increases.

Research Question Answer

```r
library(margins)
mfx <- margins(model)
summary(mfx)
```

```
 factor    AME     SE      z      p  lower  upper
    mpg 0.0470 0.0080 5.8607 0.0000 0.0313 0.0628
```

To answer our research question, what is the relationship between a car's fuel efficiency (MPG) and the probability that it has a manual transmission, for each 1 MPG increase, the probability that a car is manual increases by about 4.7 percentage points (on average).

Extending the Probit Model

When extending the probit model to include multiple predictors, the structure of the model remains unchanged:

$$P(y = 1) = \Phi(\beta_0 + \beta_1 \cdot \text{mpg} + \beta_2 \cdot \text{wt})$$

Each coefficient represents a partial effect, meaning that the effect of one variable is measured while holding the other variables constant.

If we add weight to the model, the MPG coefficient captures its effect on transmission probability while holding weight constant—meaning it reflects MPG's independent contribution rather than differences driven by weight.

Government/Politics Dataset | Implementation

Research question: to what extent do socioeconomic factors, specifically age, education, and income, predict voter turnout in American national elections?

We used a dataset provided by the American National Election Studies, detailing whether individuals voted or not in the 2024 election. The dataset had over 44 variables it considered, such as occupation and religion.

Clean Data Explanation

Translating information from the codebook, we considered the variables below.

Table 1: Variable Definitions

Variable	Name	Measurement	Coding / Description
VCF0702	Turnout	Binary	1 = Voted, 0 = Did not vote (recode)
VCF0101	Age	Continuous	Respondent age in years
VCF0110	Education	Categorical (Ordinal)	1 = LessHS, 2 = HighSchool, 3 = SomeCollege, 4 = CollegePlus
VCF0114	Income	Ordinal (treated numeric)	Higher values = higher income group
VCF0301	Party ID	Categorical (7-point scale)	StrongDem → StrongRep (7 categories)
VCF0310	Political Interest	Ordinal (1–3)	1 = Low, 2 = Medium, 3 = High interest
VCF0009z	Survey Weight	Continuous	ANES full-sample weight (post-stratification)

VCF0009z, considers if a survey accidentally talks to too many men and not enough women, the women's answers are given a higher weight (they count for more than one person) and the men's answers are given a lower weight (they count for less than one person).

We cleaned the dataset to only use the variables above

Topic 4 | Implementation

Exploratory Data Analysis

```r
chisq.test(table(df_clean$vote, df_clean$education))
```

```
	Pearson's Chi-squared test

data:  table(df_clean$vote, df_clean$education)
X-squared = 3910.4, df = 3, p-value < 2.2e-16
```

```r
chisq.test(table(df_clean$vote, df_clean$party_id))
```

```
	Pearson's Chi-squared test

data:  table(df_clean$vote, df_clean$party_id)
X-squared = 3694.5, df = 6, p-value < 2.2e-16
```

- For education, the test statistic is very large $X^2 = 3910.4$ with a p-value effectively equal to zero. We can confidently reject the idea that voting behavior is independent of education level.
- For party identification, the chi-squared test $X^2 = 3694.5$ also produces a near-zero p-value, indicating a strong relationship between party ID and turnout.

Topic 4 | Implementation

```
Warning in eval(family$initialize): non-integer #successes in a binomial glm!


Call:
glm(formula = vote ~ age + education + income + party_id + interest,
    family = binomial(link = "probit"), data = df_clean, weights = weight)

Coefficients:
                       Estimate Std. Error z value Pr(>|z|)
(Intercept)          -1.2271752  0.0323903  -37.89   <2e-16 ***
age                   0.0145019  0.0003513   41.28   <2e-16 ***
educationSomeCollege -0.2651494  0.0191872  -13.82   <2e-16 ***
educationHighSchool  -0.5935501  0.0171903  -34.53   <2e-16 ***
educationLessHS      -0.8753739  0.0229323  -38.17   <2e-16 ***
income                0.1417566  0.0047471   29.86   <2e-16 ***
party_idIndLeanDem    0.3025451  0.0226679   13.35   <2e-16 ***
party_idIndLeanRep    0.3480854  0.0235716   14.77   <2e-16 ***
party_idWeakDem       0.3703462  0.0201312   18.40   <2e-16 ***
party_idWeakRep       0.4335910  0.0221265   19.60   <2e-16 ***
party_idStrongDem     0.6847337  0.0212788   32.18   <2e-16 ***
party_idStrongRep     0.7786772  0.0244692   31.82   <2e-16 ***
interest              0.4008260  0.0075930   52.79   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 74508  on 61812  degrees of freedom
Residual deviance: 61211  on 61800  degrees of freedom
AIC: 60653

Number of Fisher Scoring iterations: 4
```

Latent scores.

Research Conclusion

```r
mfx <- margins(model1)
summary(mfx)
```

```
                factor     AME     SE        z      p   lower
                   age  0.0039 0.0001  43.1511 0.0000  0.0037
  educationHighSchool -0.1548 0.0041 -37.3483 0.0000 -0.1629
      educationLessHS -0.2449 0.0065 -37.5029 0.0000 -0.2577
 educationSomeCollege -0.0621 0.0044 -13.9750 0.0000 -0.0708
               income  0.0381 0.0013  30.1869 0.0000  0.0356
             interest  0.1076 0.0019  56.2491 0.0000  0.1039
   party_idIndLeanDem  0.0947 0.0071  13.3597 0.0000  0.0808
   party_idIndLeanRep  0.1080 0.0073  14.8308 0.0000  0.0938
    party_idStrongDem  0.1975 0.0063  31.4816 0.0000  0.1852
    party_idStrongRep  0.2192 0.0068  32.4330 0.0000  0.2059
      party_idWeakDem  0.1145 0.0063  18.1659 0.0000  0.1021
      party_idWeakRep  0.1324 0.0068  19.5564 0.0000  0.1191
```

Research Conclusion

The average marginal effect (AME) of 0.0039 means that each additional year of age increases the probability of voting by about **0.4 percentage points**, holding other factors constant. This suggests that older individuals are more likely to vote.

Compared to individuals with a college degree (the baseline group), those with lower levels of education are significantly less likely to vote. For example, having only a high school education decreases the probability of voting by about **15.5 percentage points**

The AME of 0.0381 shows that moving up one income category increases the probability of voting by about **3.8 percentage points**.

Compared to independents, individuals who identify with a political party, especially strong partisans, are substantially more likely to vote. For example, strong Democrats and strong Republicans are about **20–22 percentage points more likely** to vote than independents.

Similarly, **political interest** has one of the largest effects in the model. The average marginal effect of 0.1076 indicates that moving up one level of interest increases the probability of voting by about **10.8 percentage points**.

Overall, while socioeconomic resources are important, these results show that political engagement and partisan identity are equally powerful predictors of turnout.

Topic 5 | Comparison to Other Models

Motivation

Several methods exist for modeling binary outcomes. This section compares probit regression against three alternatives:

1. **Linear Regression (OLS)** — the standard baseline for continuous outcomes
2. **Logistic Regression** — the most widely used GLM for binary data
3. **Lasso Regression** — a penalized method that performs automatic variable selection

The goal is to understand the practical and theoretical differences between these approaches, and to identify when probit regression is the appropriate choice.

Data: Donner Party Survival

We use the **Donner Party survival dataset** from the `Sleuth3` package. The binary outcome is survival status (1 = Survived, 0 = Died), with Age and Sex as predictors.

```r
library(Sleuth3); library(dplyr); library(ggplot2)
library(glmnet);  library(caret)

set.seed(42)

# recode response and sex
donner <- case2001 %>%
  mutate(
    mystatus = if_else(Status == "Survived", 1, 0),
    male     = if_else(Sex == "Male", 1, 0)
  )

head(donner[, c("Age", "Sex", "Status", "mystatus")], 4)
```

Data Preview

```
  Age    Sex   Status mystatus
1  23   Male     Died        0
2  40 Female Survived        1
3  40   Male Survived        1
4  30   Male     Died        0
```

Linear Regression on a Binary Response

OLS fits:

$$\hat{E}(Y \mid \mathbf{X}) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_p X_p$$

When $Y \in \{0, 1\}$, this is the **Linear Probability Model (LPM)**. Since $\mathbf{X} \in \mathbb{R}$, fitted values are not constrained to $[0, 1]$, which means the model can produce predicted probabilities below 0 or above 1. This violates the basic requirement for a valid probability model and is the primary motivation for using a link function in GLMs.

OLS on a Binary Response — Code

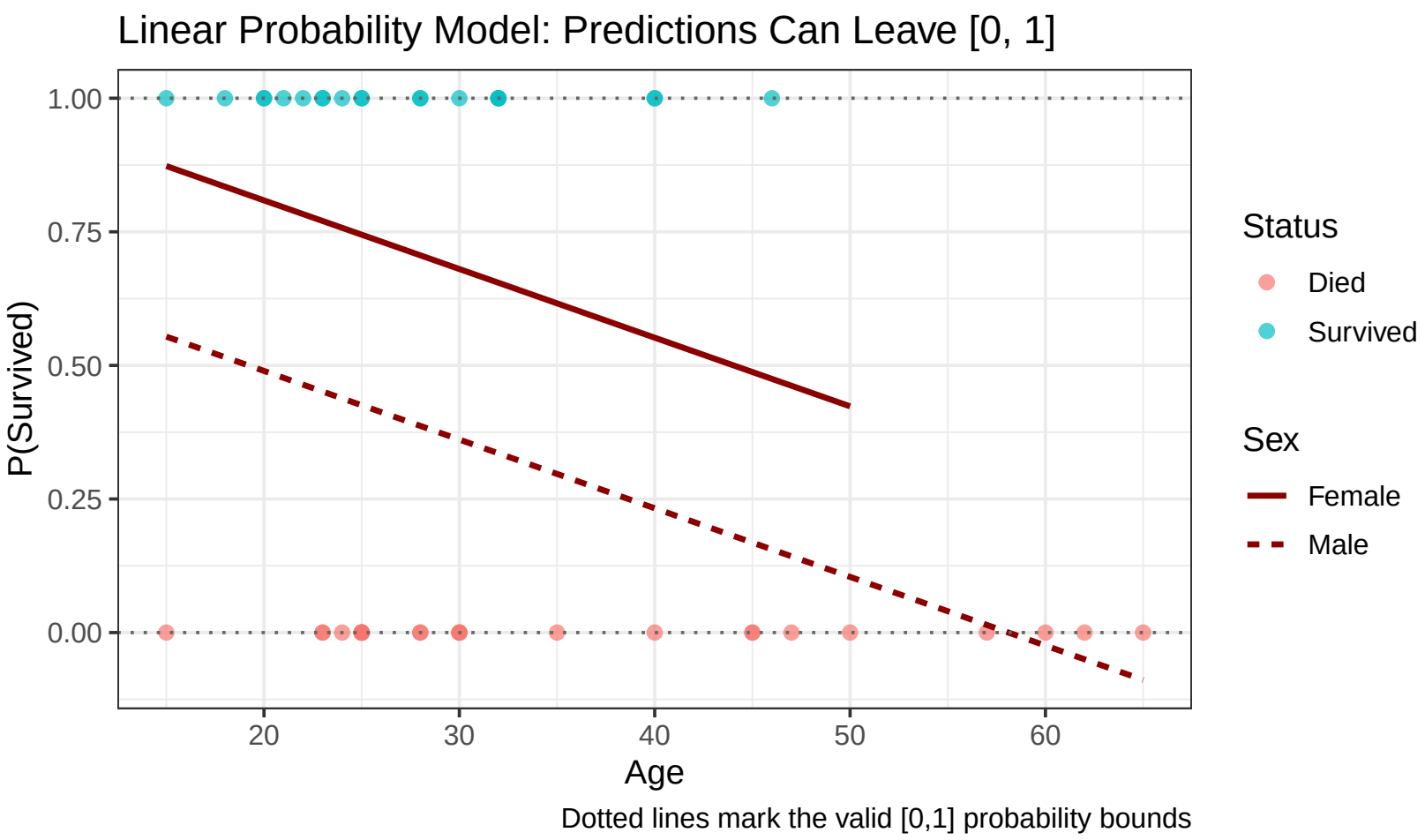
```r
# fit OLS treating the binary outcome as continuous
lpm_fit <- lm(mystatus ~ Age + male, data = donner)

# store fitted values as predicted probabilities (not bounded to [0,1])
donner <- donner %>%
  mutate(pi_lpm = predict(lpm_fit))
```

OLS on a Binary Response — Plot Code

```
ggplot(donner, aes(x = Age, y = mystatus)) +
  geom_point(aes(color = Status), size = 2, alpha = 0.7) +
  geom_line(aes(y = pi_lpm, group = male,
                linetype = factor(male, labels = c("Female","Male"))),
            color = "darkred", linewidth = 1) +
  geom_hline(yintercept = c(0, 1), linetype = "dotted", color = "gray40") +
  labs(title = "Linear Probability Model: Predictions Can Leave [0, 1]",
       y = "P(Survived)", x = "Age", linetype = "Sex",
       caption = "Dotted lines mark the valid [0,1] probability bounds") +
  theme_bw(base_size = 12)
```

OLS on a Binary Response — Plot



OLS Predictions Outside [0, 1]

```r
lpm_preds <- predict(lpm_fit)

cat("Range of OLS predictions:", round(range(lpm_preds), 4), "\n")
```

```
Range of OLS predictions: -0.0885 0.8731
```

```r
cat("Predictions below 0:     ", sum(lpm_preds < 0), "\n")
```

```
Predictions below 0:      3
```

```r
cat("Predictions above 1:     ", sum(lpm_preds > 1), "\n")
```

```
Predictions above 1:      0
```

```r
# misclassification at 0.5 threshold
lpm_class <- ifelse(lpm_preds > 0.5, 1, 0)
cat("OLS misclassification rate:",
    round(mean(lpm_class != donner$mystatus), 4), "\n")
```

```
OLS misclassification rate: 0.2444
```

Even on this small dataset, OLS produces invalid probabilities. Linear regression is not appropriate for binary outcomes.

Logistic Regression

Logistic regression uses the **logit link**:

$$\eta = \text{logit}(\pi) = \ln\left[\frac{\pi}{1-\pi}\right] = \mathbf{x}^T\beta \quad \Rightarrow \quad \hat{\pi} = \frac{e^\eta}{1+e^\eta}$$

This assumes errors follow a logistic distribution.
Probit regression uses the **inverse standard normal CDF**:

$$\eta = \Phi^{-1}(\pi) = \mathbf{x}^T\beta \quad \Rightarrow \quad \hat{\pi} = \Phi(\mathbf{x}^T\hat{\beta})$$

This assumes errors follow a standard normal distribution. Both are valid GLMs for binary outcomes — they differ only in their distributional assumption.

Link Functions — Code

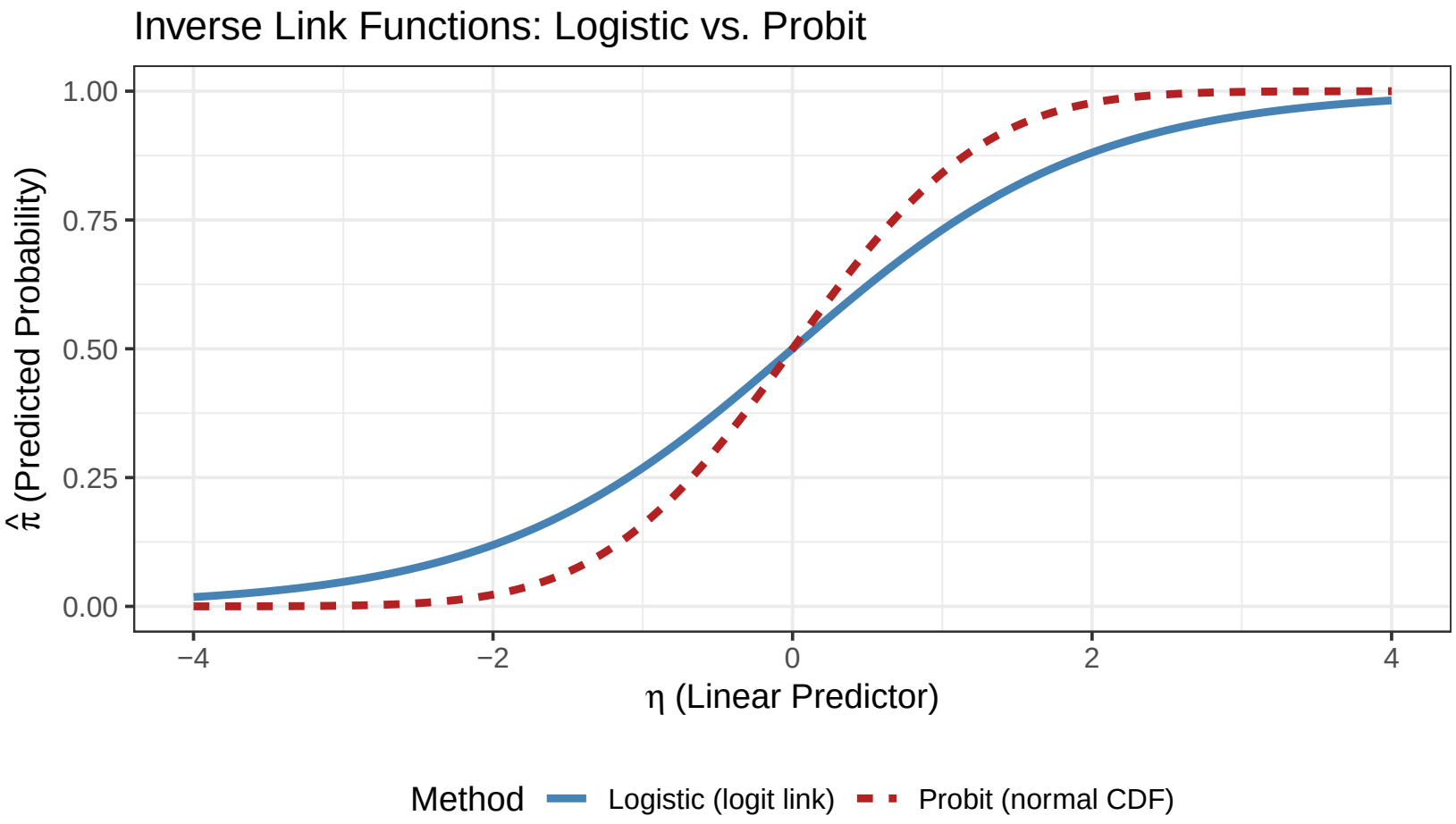
```r
eta_seq <- seq(-4, 4, length.out = 300)

df_links <- data.frame(
  eta    = rep(eta_seq, 2),
  pi     = c(plogis(eta_seq),  # inverse logit
             pnorm(eta_seq)),  # inverse probit (std normal CDF)
  Method = rep(c("Logistic (logit link)", "Probit (normal CDF)"), each = 300)
)
```

Link Functions — Plot Code

```
ggplot(df_links, aes(x = eta, y = pi, color = Method, linetype = Method)) +
  geom_line(linewidth = 1.3) +
  labs(title = "Inverse Link Functions: Logistic vs. Probit",
       x = expression(eta ~ "(Linear Predictor)"),
       y = expression(hat(pi) ~ "(Predicted Probability)")) +
  scale_color_manual(values = c("steelblue", "firebrick")) +
  theme_bw(base_size = 12) + theme(legend.position = "bottom")
```

Link Functions — Plot



The probit curve has lighter tails — it approaches 0 and 1 faster than the logistic curve.

Fitting Both Models — Code

```r
# fit logistic and probit
logit_fit  <- glm(mystatus ~ Age + male, data = donner,
                  family = binomial(link = "logit"))

probit_fit <- glm(mystatus ~ Age + male, data = donner,
                  family = binomial(link = "probit"))

# compare coefficients
round(cbind(
  "OLS Coef"    = coef(lpm_fit),
  "Logit Coef"  = coef(logit_fit),
  "Probit Coef" = coef(probit_fit)
), 4)
```

Fitting Both Models — Output

	OLS Coef	Logit Coef	Probit Coef
(Intercept)	1.0658	3.2304	1.9173
Age	-0.0128	-0.0782	-0.0457
male	-0.3192	-1.5973	-0.9583

The probit coefficients are smaller by a factor of roughly 1.6 to 1.8. The logistic distribution has variance $\pi^2/3 \approx 3.29$, while the standard normal has variance 1. The scales differ, but the substantive conclusions are the same.

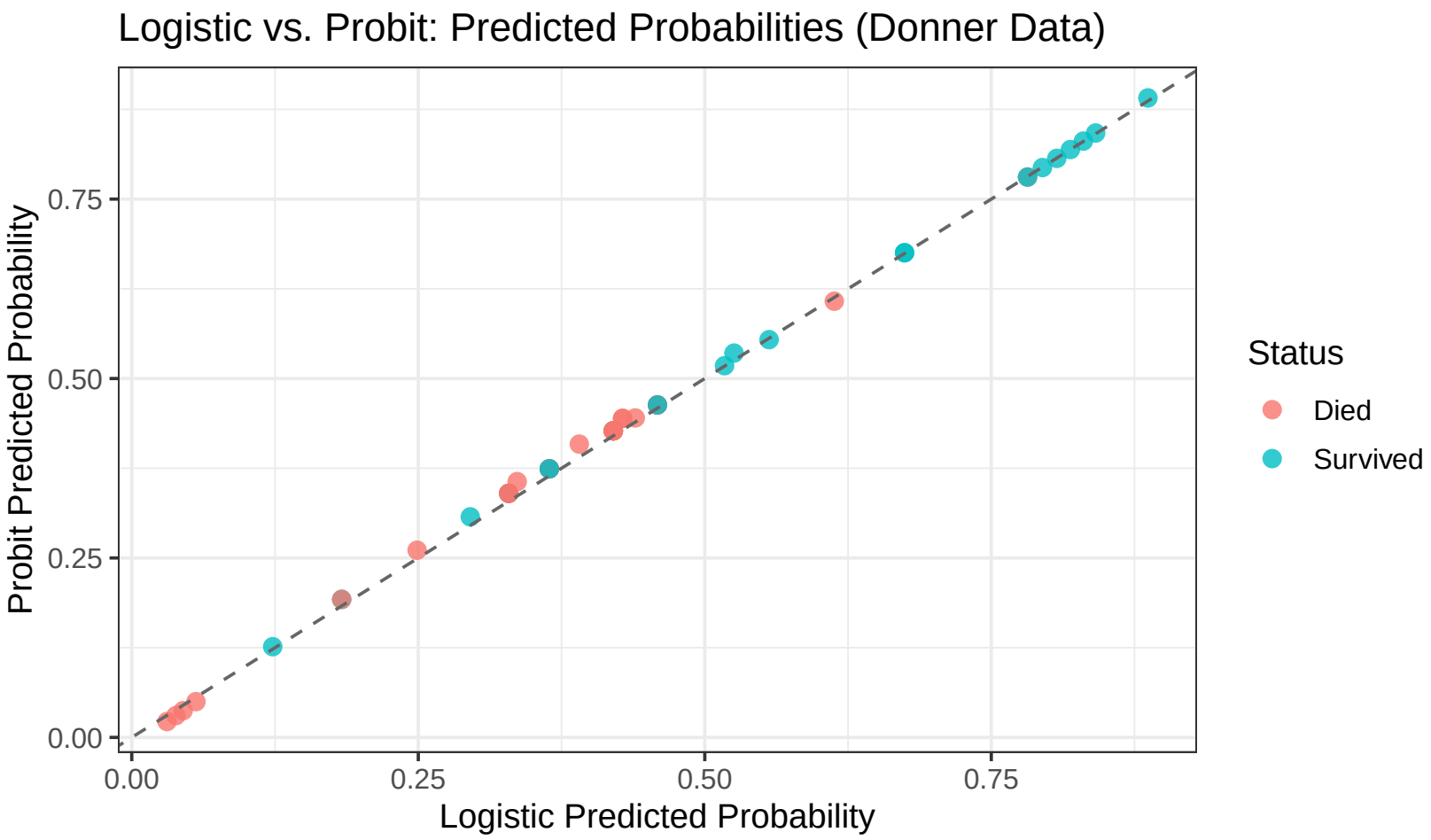
Predicted Probabilities — Code

```
donner <- donner %>%
  mutate(
    pi_logit  = predict(logit_fit,  type = "response"),
    pi_probit = predict(probit_fit, type = "response")
  )
```

Predicted Probabilities — Plot Code

```r
ggplot(donner, aes(x = pi_logit, y = pi_probit, color = Status)) +
  geom_point(size = 2.5, alpha = 0.8) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray40") +
  labs(title = "Logistic vs. Probit: Predicted Probabilities (Donner Data)",
       x = "Logistic Predicted Probability",
       y = "Probit Predicted Probability") +
  theme_bw(base_size = 12)
```

Predicted Probabilities — Plot



The predicted probabilities from both models are nearly identical, confirming that the choice of link function has minimal practical impact on fitted values.

Model Fit Metrics — Code

```r
cat(sprintf("%-28s %8s %8s\n", "Metric", "Logistic", "Probit"))
cat(sprintf("%-28s %8.3f %8.3f\n", "AIC",
            AIC(logit_fit), AIC(probit_fit)))
cat(sprintf("%-28s %8.3f %8.3f\n", "Residual Deviance",
            deviance(logit_fit), deviance(probit_fit)))
cat(sprintf("%-28s %8.3f %8.3f\n", "Null Deviance",
            logit_fit$null.deviance, probit_fit$null.deviance))

log_err  <- mean((donner$pi_logit  > 0.5) != donner$mystatus)
prob_err <- mean((donner$pi_probit > 0.5) != donner$mystatus)
cat(sprintf("%-28s %8.4f %8.4f\n", "Misclassification Rate",
            log_err, prob_err))
```

Model Fit Metrics — Output

Metric	Logistic	Probit
AIC	57.256	57.283
Residual Deviance	51.256	51.283
Null Deviance	61.827	61.827
Misclassification Rate	0.2222	0.2222

The two models produce nearly identical fit across every metric. AIC differs by less than 0.03, and misclassification rates are equal. The choice between logistic and probit is rarely driven by model performance — it is almost always a question of convention, interpretability, or the underlying distributional assumption.

When to Prefer Probit

1. Latent variable framework. Assume an unobserved variable $Y^* = \mathbf{x}^T\beta + \varepsilon$ where $\varepsilon \sim N(0,1)$. We observe $Y = 1$ if $Y^* > 0$. This is the probit model exactly — a standard framework in economics and political science.

2. Tail behavior. The normal CDF approaches 0 and 1 more rapidly than the logistic distribution. Probit is preferable when probabilities near the extremes are of scientific interest.

3. Field convention. Probit is the standard choice in economics, political science, and related disciplines. For government and political science applications, probit is the conventional model.

4. Marginal effects. Probit coefficients do not have an odds-ratio interpretation, but average marginal effects provide a clear and directly interpretable quantity.

Marginal Effects

Probit coefficients have no direct probability interpretation on their own. The standard approach is to compute **Average Marginal Effects (AME)**:

$$\text{AME}_j = \frac{1}{n} \sum_{i=1}^{n} \phi(\mathbf{x}_i^T \hat{\boldsymbol{\beta}}) \cdot \hat{\beta}_j$$

where $\phi(\cdot)$ is the standard normal PDF.

```r
library(margins)

# compute AMEs
ame <- margins(probit_fit)
summary(ame)[, c("factor", "AME", "SE", "z", "p")]
```

```
 factor     AME     SE       z      p
    Age -0.0148 0.0057 -2.6113 0.0090
   male -0.3109 0.1188 -2.6160 0.0089
```

This gives the change in predicted probability for a one-unit change in each predictor, averaged across all observations.

Lasso Regression

The logistic or probit objective function can be **penalized** to perform variable selection:

$$\hat{\boldsymbol{\beta}} = \text{argmin}_{\boldsymbol{\beta}} \left\{ -\frac{2}{n}\ell(\boldsymbol{\beta}) + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

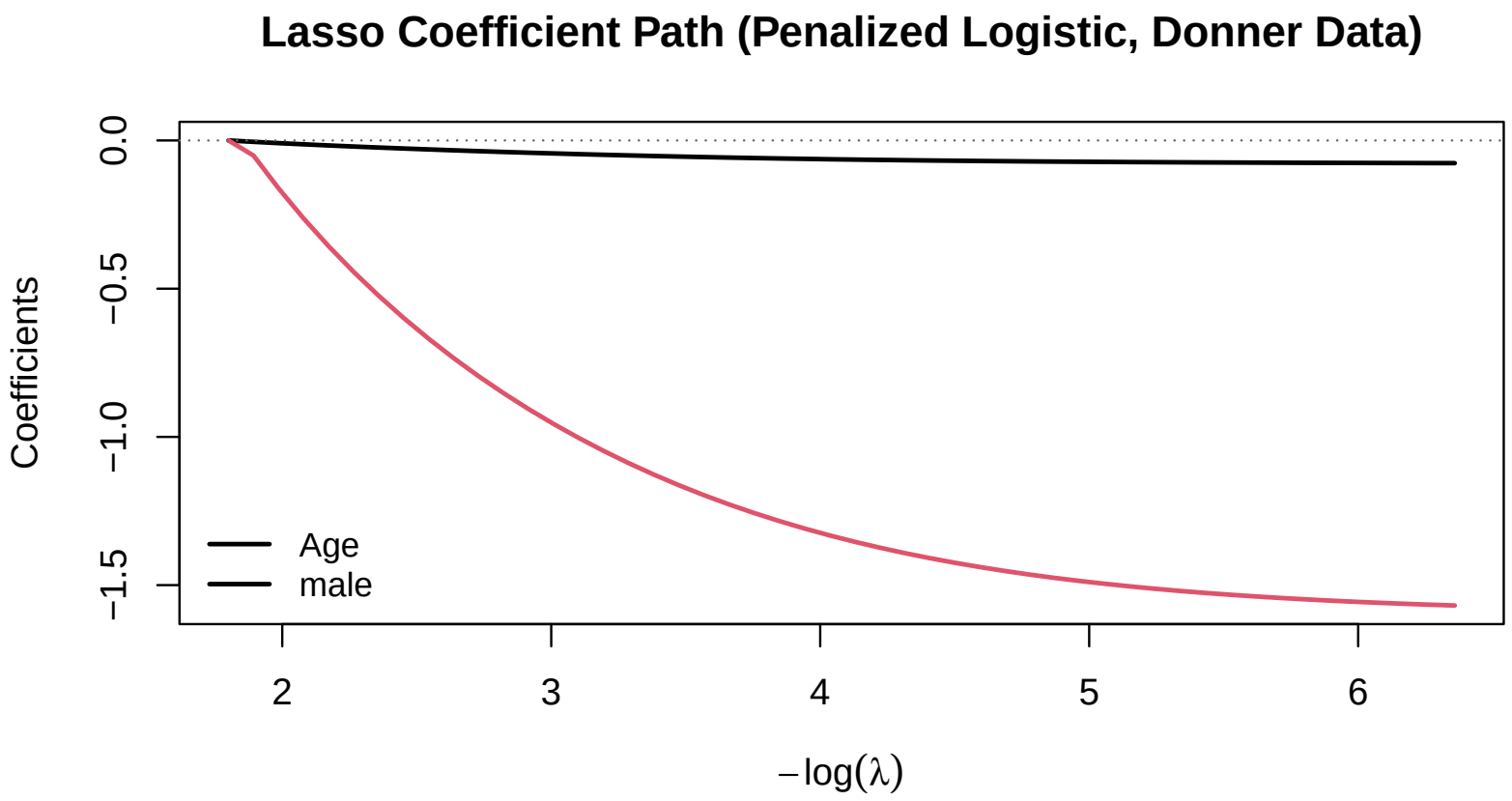
The ℓ_1 penalty shrinks small coefficients toward zero and eliminates irrelevant predictors entirely when λ is large enough. This is particularly useful in high-dimensional settings such as political science data with many demographic and behavioral predictors.

Lasso: Fitting the Model — Code

```r
# glmnet requires a plain matrix, not a model formula
X_mat <- model.matrix(mystatus ~ Age + male, data = donner)[, -1]
y_vec <- donner$mystatus

# fit lasso path over 100 values of lambda (alpha=1 selects lasso over ridge)
lasso_path <- glmnet(X_mat, y_vec, family = "binomial", alpha = 1)
```

Lasso: Coefficient Path



As λ increases (moving left on the x-axis), both coefficients shrink toward zero. The `male` coefficient shrinks more slowly, indicating it is the stronger predictor of survival. `Age` enters the model quickly but with a smaller effect.

Lambda Selection via Cross-Validation — Code

```r
set.seed(42)
# 10-fold CV to select lambda; type.measure="class" minimizes misclassification
cv_lasso <- cv.glmnet(X_mat, y_vec,
                      family       = "binomial",
                      alpha        = 1,
                      nfolds       = 10,
                      type.measure = "class")

# lambda.min: lowest CV error; lambda.1se: most regularized model within 1 SE
cat("Lambda (min CV error): ", round(cv_lasso$lambda.min, 5), "\n")
```

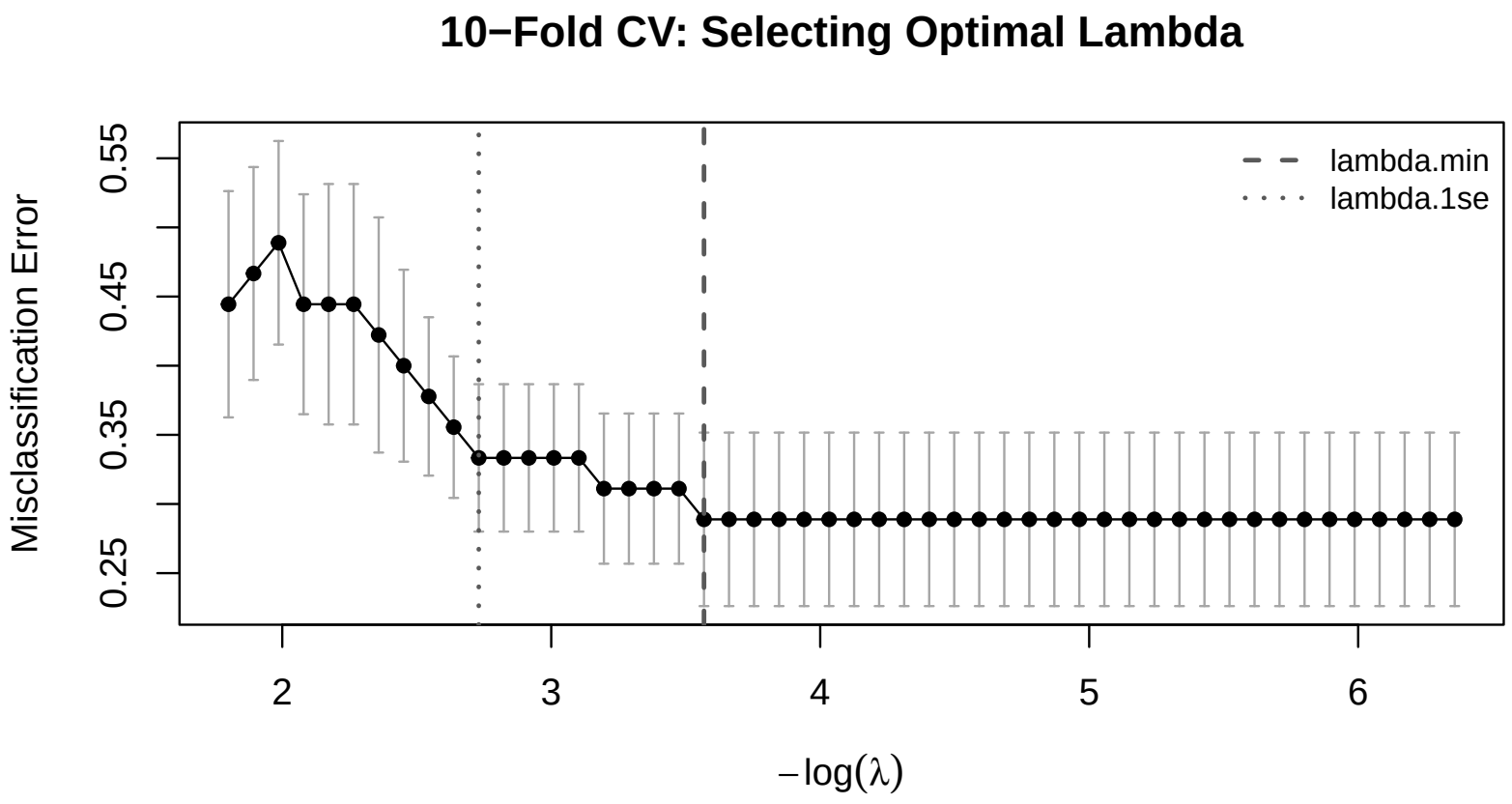
```
Lambda (min CV error):  0.02822
```

```r
cat("Lambda (1 SE rule):    ", round(cv_lasso$lambda.1se, 5), "\n")
```

```
Lambda (1 SE rule):     0.06518
```

Lambda Selection via Cross-Validation — Plot

10-fold CV selects λ_{min} as the value minimizing misclassification error, and λ_{1SE} as the largest λ within one standard error of that minimum — the more parsimonious choice. Error stabilizes well before the full-model limit, indicating that a small penalty is sufficient here.



Lasso Final Coefficients

```r
cat("Lasso coefficients at lambda.min:\n")
```

```
Lasso coefficients at lambda.min:
```

```r
round(as.matrix(coef(cv_lasso, s = "lambda.min")), 5)
```

```
            lambda.min
(Intercept)    2.30441
Age           -0.05615
male          -1.19653
```

```r
cat("\nLasso coefficients at lambda.1se:\n")
```

```

Lasso coefficients at lambda.1se:
```

```r
round(as.matrix(coef(cv_lasso, s = "lambda.1se")), 5)
```

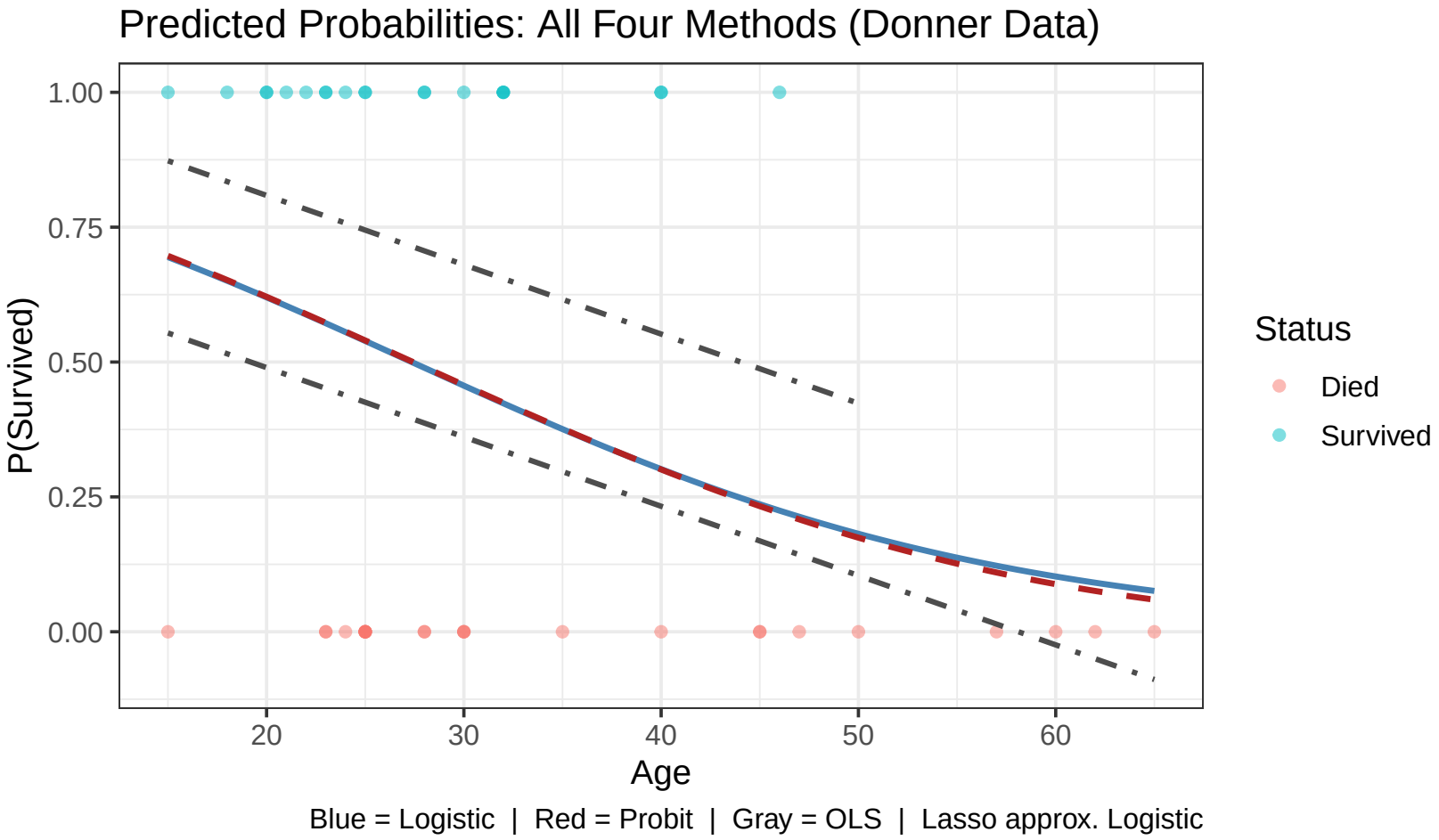
```
            lambda.1se
(Intercept)    1.43295
Age           -0.03615
male          -0.79566
```

A dot (.) indicates a coefficient shrunk exactly to zero — that predictor has been

All Four Methods — Code

```r
ggplot(donner, aes(x = Age)) +
  geom_point(aes(y = mystatus, color = Status), alpha = 0.5, size = 1.5) +
  geom_line(aes(y = pi_lpm, group = male), color = "gray30",
            linewidth = 0.9, linetype = "dotdash") +
  geom_smooth(aes(y = mystatus), method = "glm",
              method.args = list(family = binomial(link = "logit")),
              se = FALSE, color = "steelblue", linewidth = 1) +
  geom_smooth(aes(y = mystatus), method = "glm",
              method.args = list(family = binomial(link = "probit")),
              se = FALSE, color = "firebrick", linewidth = 1, linetype = "dashed") +
  labs(title = "Predicted Probabilities: All Four Methods (Donner Data)",
       y = "P(Survived)", x = "Age",
       caption = "Blue = Logistic  |  Red = Probit  |  Gray = OLS  |  Lasso approx. Logistic") +
  theme_bw(base_size = 12)
```

All Four Methods — Plot



Confusion Matrix Comparison — Code

```r
# classify each observation using a 0.5 probability threshold
threshold <- 0.5

results <- donner %>%
  mutate(
    # convert predicted probabilities to binary class labels
    pred_lpm    = factor(ifelse(pi_lpm    > threshold, 1, 0)),
    pred_logit  = factor(ifelse(pi_logit  > threshold, 1, 0)),
    pred_probit = factor(ifelse(pi_probit > threshold, 1, 0)),
    # lasso: get response-scale predictions at the CV-selected lambda
    pred_lasso  = factor(ifelse(
      predict(cv_lasso, newx = X_mat, s = "lambda.min",
              type = "response")[, 1] > threshold, 1, 0)),
    actual = factor(mystatus)
  )
```

Confusion Matrix Comparison — Accuracy Code

```r
# helper: proportion of correctly classified observations
acc <- function(pred, actual) round(mean(pred == actual), 4)

cat(sprintf("%-20s %s\n", "Method",    "Accuracy"))
cat(sprintf("%-20s %s\n", "OLS (LPM)", acc(results$pred_lpm,    results$actual)))
cat(sprintf("%-20s %s\n", "Logistic",  acc(results$pred_logit,  results$actual)))
cat(sprintf("%-20s %s\n", "Probit",    acc(results$pred_probit, results$actual)))
cat(sprintf("%-20s %s\n", "Lasso",     acc(results$pred_lasso,  results$actual)))
```

Confusion Matrix Comparison — Output

```
Method                Accuracy

OLS (LPM)             0.7556

Logistic              0.7778

Probit                0.7778

Lasso                 0.7556
```

Logistic and probit tie at 77.78%, while OLS and lasso both fall to 75.56%. The bounded probability models classify slightly better here, though on a dataset of 45 observations the difference is small. Probit and logistic are effectively equivalent in predictive accuracy on this data.

Probit Confusion Matrix

```
          Reference
Prediction  0  1
         0 23  8
         1  2 12

Accuracy             : 0.7778
Sensitivity          : 0.6000
Specificity          : 0.9200
Pos Pred Value       : 0.8571
Neg Pred Value       : 0.7419
Kappa                : NA
```

The probit model is considerably better at identifying deaths (specificity = 0.92) than survivors (sensitivity = 0.60). In the Donner Party data, older males had substantially lower survival odds, and the model captures that pattern well at the majority class. The imbalance between sensitivity and specificity is typical when the outcome is unequally distributed.

Model Comparison Summary

Criterion	OLS (LPM)	Logistic	Probit	Lasso
Predictions in [0, 1]?	No	Yes	Yes	Yes
Link function	Identity	Logit	Φ^{-1} (normal)	Logit or Probit
Error distribution	Normal	Logistic	Normal	Logistic
Estimation method	OLS (closed form)	MLE (IRLS)	MLE (Fisher Scoring)	MLE + ℓ_1 penalty

Model Comparison Summary

Criterion	OLS (LPM)	Logistic	Probit	Lasso
Coefficient interpretation	$\Delta\hat{y}$	Log-odds / Odds ratio	Marginal effects (AME)	Log-odds (shrunk)
Variable selection?	No	No	No	Yes
Tuning required?	No	No	No	Yes (λ)
Preferred when...	Never	General / odds needed	Econ, politics, latent variable	Many predictors
Inference (p-values, CIs)?	Yes	Yes	Yes	Not standard

Key Takeaways

- **Linear regression (OLS)** is not appropriate for binary outcomes. Predicted values are not bounded to $[0, 1]$ and the model is misspecified.
- **Logistic and probit regression** produce nearly identical predicted probabilities and model fit. Coefficients differ in scale by approximately 1.6x due to differing error distributions, but substantive conclusions are the same.
- **Probit is preferred** in economics and political science due to the latent variable framework and field convention. Coefficients are interpreted via average marginal effects.
- **Logistic regression is preferred** when odds ratios are the natural effect measure.
- **Lasso** extends either model with an ℓ_1 penalty for automatic variable selection. Standard p-values and confidence intervals are not available after lasso selection.
- For government and political science applications, probit regression is the standard and appropriate choice.

References

Citations

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Packages

- R Packages: `glmnet`, `caret`, `margins`, `Sleuth3`, `dplyr`, `ggplot2`

Appendix

```r
model <- glm(am ~ mpg,
             data = mtcars,
             family = binomial(link = "probit"))

mpg_seq <- seq(min(mtcars$mpg), max(mtcars$mpg), length.out = 100)

pred_prob <- predict(model, newdata = data.frame(mpg = mpg_seq), type = "response")

plot(mtcars$mpg, mtcars$am,
     xlab = "MPG",
     ylab = "Probability of Manual (am = 1)",
     main = "Probit Regression: MPG vs Manual Transmission",
     pch = 16)

lines(mpg_seq, pred_prob, lwd = 2)
```

```r
df <- read.csv("../anes_cleaned.csv", stringsAsFactors = FALSE)


vars <- c("VCF0702",  # turnout
          "VCF0101",  # age
          "VCF0110",  # education
          "VCF0114",  # income
          "VCF0301",  # party ID
          "VCF0310",  # political interest
          "VCF0009z") # weights

df_clean <- df[, vars]

df_clean[df_clean < 0] <- NA

# Dependent Variable: Turnout (1 = voted, 0 = did not vote)
df_clean$vote <- ifelse(df_clean$VCF0702 == 2, 1,
                  ifelse(df_clean$VCF0702 == 1, 0, NA))

df_clean$age <- as.numeric(df_clean$VCF0101)

df_clean$education <- factor(df_clean$VCF0110,
                             levels = c(4,3,2,1),
                             labels = c("CollegePlus", "SomeCollege", "HighSchool", "LessHS"))

df_clean$income <- as.numeric(df_clean$VCF0114)

df_clean$party_id <- factor(df_clean$VCF0301,
                            levels = c(4,3,5,2,6,1,7),
                            labels = c("Independent",
                                       "IndLeanDem",
                                       "IndLeanRep",
                                       "WeakDem",
                                       "WeakRep",
                                       "StrongDem",
                                       "StrongRep"))

df_clean$interest <- as.numeric(df_clean$VCF0310)

df_clean$weight <- df_clean$VCF0009z


df_clean <- na.omit(df_clean)
```


```r
model <- glm(am ~ mpg,
             data = mtcars,
             family = binomial(link = "probit"))

mpg_seq <- seq(min(mtcars$mpg), max(mtcars$mpg), length.out = 100)

pred_prob <- predict(model, newdata = data.frame(mpg = mpg_seq), type = "response")

plot(mtcars$mpg, mtcars$am,
     xlab = "MPG",
     ylab = "Probability of Manual (am = 1)",
     main = "Probit Regression: MPG vs Manual Transmission",
     pch = 16)

lines(mpg_seq, pred_prob, lwd = 2)
```


- Probit model
- Maximum Likelihood details
- Convergence tolerance pseudocode logic